

Max Tong

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EDUCATION

Northwestern University

Expected Jun 2027

Bachelor of Science in Computer Science - Minor in Math - Major GPA: 3.7/4.0

Evanston, IL

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems (C/C++), Artificial Intelligence, Machine Learning, Scalable Software (AWS), Networking (TCP/UDP)
- **Clubs & Organizations:** Northwestern Garage Residency, Formula SAE, Competitive Programming (ICPC)

WORK EXPERIENCE

AT&T

Atlanta, GA

Software Engineering Intern

Jun 2026 - Present

- Engineered a Random Forest and LightGBM fraud detection model achieving **0.90 AUC** in replicating Visa Decision Manager outputs, enabling **80%** reduction in third-party API calls for Cricket BYOD eSIM activations
- Performed EDA and feature engineering on MongoDB fraud data using Pandas & deployed model via Databricks
- Implemented a Redis caching Python + FastAPI microservice for fraud velocity and morphing data, screening **10,000+** daily transactions at sub-250ms p95 latency to flag fraud before impacting customers.

TwoCann

Evanston, IL

Founder

Jun 2024 - Present

- Built, shipped, and scaled a cross-platform consumer app to **1,000+** registered users on **Firebase** and **Google Cloud Platform** as sole developer.
- Wrote native **Swift** code for iOS widgets alongside **Dart** and **JavaScript**; designed efficient data models and caching to reduce Firestore reads/writes by **30%** and integrated Firebase FCM tokens for push notifications.
- Built and debugged web service (HTTPS callable functions), utilized GCP logging/metrics to fix reliability issues.
- Led a team in Northwestern's startup incubator (The Garage); shipped features using Git version control.

Clemson University

Clemson, SC

Research Assistant

Jun 2025 - Sep 2025

- Developed a serverless web app with **React/Node.js** for user check-in for lab machinery. Used **AWS RDS** to track user sessions and machine usage across shared equipment & **S3** to store lab setting snapshots & survey data
- Implemented a **CI/CD** pipeline using GitHub Actions to automatically build, test, and deploy after every commit
- Built and deployed a Hugo static website used by **200+** internal users to access documentation and lab resources.
- Debugged cross platform runtime issues in a **Linux Docker** container, identifying several OS-dependent bugs.

RESEARCH

Prescience Lab (Prof. Peter Dinda) | C/C++, RDMA, InfiniBand

Dec 2025 - Present

- Designed an RDMA gather collective in **C/C++** over **libibverbs**, translating a tree-based MPI gather into a triggered architecture that offloads the gather to the NIC and removes the CPU from the critical path.
- Reduced per-node operations from a 3-step (RDMA write, atomic fetch-and-add, doorbell write) sequence by exploiting InfiniBand within-QP ordering guarantees to eliminate explicit memory fences.
- Engineered a **kernel-bypass** networking flow for direct NIC-to-memory communication across supercomputing-cluster nodes, using MPI as an out-of-band bootstrap channel.

PROJECTS

Reliable Transport Protocol from Scratch | Python, Concurrency, Networking

Dec 2025 - March 2026

- Engineered a copy of TCP, simulating packet loss, data corruption, and ensuring reliable delivery through sequence numbering, MD5 packet hashing, and timeout based packet retransmission.
- Maximized network throughput by implementing pipelined ACKs and background listener threads.

Test Driven Monopoly Game | Java, JUnit 5, EasyMock, GitHub Actions

March 2026 - Jun 2026

- Built a full Java Monopoly engine with a team using **TDD** and **BVA**; achieved **100%** branch/line coverage (JaCoCo) with mutation testing (PITest) and automated CI/CD regression on every commit.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Rust, JavaScript/TypeScript, SQL (Postgres), Swift, Go

Frameworks & Stack: React, Express, Flutter, Flask, PyTorch, TensorFlow, AWS, Kubernetes, Firebase, Docker

Systems & Tools: Linux, Git, CI/CD, RDMA/InfiniBand, libibverbs, Docker, Backend systems/APIs